

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0104

Rio Dell 138 kV
January 22, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0104. The Interconnection Customer proposes a 179.3 MW wind facility interconnecting at the Rio Dell 138 kV substation in the West Virginia area. The request was received on February 13, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 92 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,194,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The RIO DELL SUBSTATION to CELERON 138kV is overloaded to 142.3% of its 221 MVA emergency rating for the loss of the MONTANO SUBSTATION to RIO DELL SUBSTATION 138kV Line. The project contributes 68.8% DFAX (123.3 MW) to this overload.

For the loss of CELERN to RIO DELL 138kV, the RIO DELL to MONTANO 138kV loads to 137.0% of its 242 MVA rating. Project DFAX: 72.5%.

The MONTANO to CELERON 138kV is overloaded to 136.1% of its 241 MVA emergency rating for the loss of the CELERON to RIO DELL 138kV. The project contributes 72.5% DFAX (129.9 MW) to this overload.

The CELERON to SANDALFOOT SUBSTATION 138kV is overloaded to 107.9% of its 228 MVA emergency rating for the loss of the CELERON SUBSTATION to YELM PP 138kV. The project contributes 51.1% DFAX (91.7 MW) to this overload.

SANDALFOOT to N MESA 138kV: 106.1% loading (219 MVA) following the outage of the CELERON SUBSTATION to YELMPP 138kV; DFAX = 51.1%.

CELERON SUBSTATION to YELM PP 138kV line: 102.4% loading (242 MVA) following the outage of the CELERON to SANDALFOOT 138kV; DFAX = 49.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RIO DELL SUBSTATION to CELERON 138kV	Rebuild line with higher-capacity conductor	\$4,620,000	100% (\$4,620,000)
RIODELL to MONTANO 138kV	Rebuild line with higher-capacity conductor	\$2,273,000	100% (\$2,273,000)
MONTANO to CLERON 138kV	Rebuild line with higher-capacity conductor	\$1,346,000	100% (\$1,346,000)
CELERON to SANDALFOOT SUBSTATION 138kV	Reconductor line	\$1,607,000	100% (\$1,607,000)
SANDALFOOT to N MESA 138kV	Reconductor line	\$629,000	100% (\$629,000)
CELERON to YELM PP 138kV	Reconductor line	\$2,719,000	100% (\$2,719,000)
	Total allocated to project		\$13,194,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.